

COSC 594

LLM Alignment (IV)

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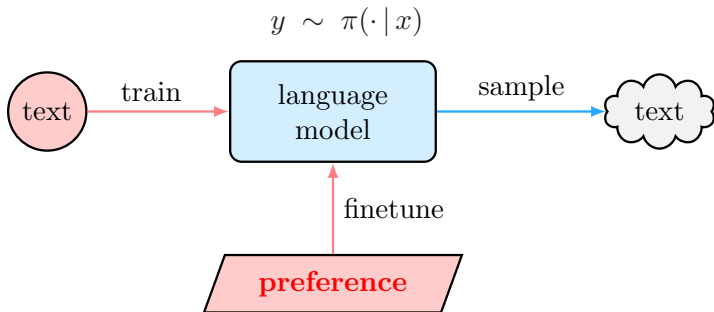
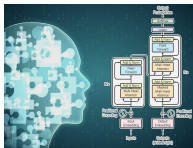
Outline

- RECAP
- GENERAL PREFERENCE
- NASH RLHF

Outline

■ RECAP

LLM alignment framework



preference – reward / human value

Preference

SFT policy : $\pi_{\text{ref}}(\cdot | x) : X \rightarrow \Delta_Y$

Response: $y \sim \pi_{\text{ref}}(\cdot | x)$

Preference: $y_1 \succ y_0 | x$

■ BRADLEY-TERRY MODEL

$$\mathbb{P}(y_1 \succ y_0 | x) = \frac{e^{r(x, y_1)}}{e^{r(x, y_1)} + e^{r(x, y_0)}} := \sigma(r(x, y_1) - r(x, y_0))$$

$$\sigma(t) = 1/(1 + e^{-t})$$

Consistency between reward and preference

A counterexample

Preference #1: $y_1 \succ y_0 \mid x$

Preference #2: $y_0 \succ y_1 \mid x$

■ BRADLEY-TERRY MODEL

$$\mathbb{P}^1(y_1 \succ y_0 \mid x) = \sigma(r(x, y_1) - r(x, y_0)) > \frac{1}{2}$$

$$\mathbb{P}^2(y_0 \succ y_1 \mid x) = \sigma(r(x, y_0) - r(x, y_1)) > \frac{1}{2}$$

$$\sigma(t) = 1/(1 + e^{-t})$$

$\Rightarrow r(x, y_1) > r(x, y_0)$ contradicts $r(x, y_0) > r(x, y_1)$

Irrationality in human preference

Multi-objective idea

Preference #1: $y_1 \succ y_0 \mid x$

Preference #2: $y_0 \succ y_1 \mid x$

■ BRADLEY-TERRY MODEL

$$\mathbb{P}^1(y_1 \succ y_0 \mid x) = \sigma(r^1(x, y_1) - r^2(x, y_0))$$

$$\mathbb{P}^2(y_0 \succ y_1 \mid x) = \sigma(r^2(x, y_0) - r^2(x, y_1))$$

$$\sigma(t) = 1/(1 + e^{-t})$$

$$r(x, y) = w^1 r^1(x, y) + w^2 r^2(x, y)$$

Unknown decoupling of preferences

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General preference

Preference oracle: $\mathbb{P}(y_1 \succ y_0 \mid x)$

Likelihood maximum : $\mathbb{P} \in \operatorname{argmax}_P \mathbb{E}_{x, y_1 \sim \pi_1, y_0 \sim \pi_0, h} \left[\log P(y_1^h \succ y_0^h \mid x) \right]$

■ WINNING PROBABILITY OF A RESPONSE

$$\mathbb{P}(y_1 \succ \pi_0 \mid x) = \mathbb{E}_{y_0 \sim \pi_0(\cdot \mid x)} [\mathbb{P}(y_1 \succ y_0 \mid x)]$$

■ WINNING PROBABILITY OF A POLICY

$$\mathbb{P}(\pi_1 \succ \pi_0 \mid x) = \mathbb{E}_{y_1 \sim \pi_1(\cdot \mid x), y_0 \sim \pi_0(\cdot \mid x)} [\mathbb{P}(y_1 \succ y_0 \mid x)]$$

Worst-case win rate

Win rate #1: $\mathbb{P}(\pi_1 \succ \pi_0)$

Win rate #2: $\mathbb{P}(\pi_0 \succ \pi_1)$

$$\mathbb{P}(\pi_1 \succ \pi_0) = \mathbb{E}_x [\mathbb{P}(\pi_1 \succ \pi_0 \mid x)]$$

$$\mathbb{P}(\pi_0 \succ \pi_1) = \mathbb{E}_x [\mathbb{P}(\pi_0 \succ \pi_1 \mid x)]$$

■ ANTISYMMETRY

$$\mathbb{P}(\pi_0 \succ \pi_1) = 1 - \mathbb{P}(\pi_1 \succ \pi_0)$$

Maximizing Win rate #1 $\equiv$ minimizing Win rate #2

Two-player game

$$\underset{\pi_1}{\text{maximize}} \quad \underset{\pi_0}{\text{minimize}} \quad \mathbb{P}(\pi_1 \succ \pi_0)$$

$$\mathbb{P}(\pi_1 \succ \pi_0) = \mathbb{E}_x [\mathbb{P}(\pi_1 \succ \pi_0 \mid x)]$$

■ BILINEAR GAME

$$\mathbb{P}(\pi_1 \succ \pi_0) = \mathbb{E}_x \mathbb{E}_{y_1 \sim \pi_1(\cdot \mid x), y_0 \sim \pi_0(\cdot \mid x)} [\mathbb{P}(y_1 \succ y_0 \mid x)]$$

$\Downarrow$

$$\underset{\pi_1}{\text{maximize}} \underset{\pi_0}{\text{minimize}} \mathbb{P}(\pi_1 \succ \pi_0) = \underset{\pi_0}{\text{minimize}} \underset{\pi_1}{\text{maximize}} \mathbb{P}(\pi_1 \succ \pi_0)$$

Existence of Nash equilibria & exchangeable minimax order

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Mirror descent self-play

$$\pi^{(t+1)}(\cdot | x) = \operatorname{argmax}_{\pi(\cdot | x)} \mathbb{E}_{y \sim \pi(\cdot | x)} \left[\mathbb{P}(y \succ \pi^{(t)} | x) \right] - \frac{1}{\eta} D_{\text{KL}}(\pi(\cdot | x) \| \pi^{(t)}(\cdot | x))$$

$\eta > 0$ – stepsize

★ explicit update

$$\pi^{(t+1)}(y | x) = \pi^{(t)}(y | x) e^{\eta \mathbb{P}(y \succ \pi^{(t)} | x)} \frac{1}{Z^{(t)}(x)}$$

$$\text{or } \eta \mathbb{P}(y \succ \pi^{(t)} | x) = \log \frac{\pi^{(t+1)}(y | x)}{\pi^{(t)}(y | x)} + \log Z^{(t)}(x)$$

$$Z^{(t)}(x) = \sum_y \pi^{(t)}(y | x) e^{\eta \mathbb{P}(y \succ \pi^{(t)} | x)} - \text{normalization constant}$$

Relation between preference and target policy

Nonlinear regression

$$\pi^{(t+1)}(\cdot | x) = \underset{\pi(\cdot | x)}{\operatorname{argmin}} \mathbb{E}_{x, y \sim \pi^{(t)}(\cdot | x)} \left(\log \frac{\pi(y | x)}{\pi^{(t)}(y | x)} - \left(\eta \mathbb{P}(y \succ \pi^{(t)} | x) - \log Z^{(t)}(x) \right) \right)^2$$

$\eta > 0$ – stepsize

★ empirical estimates

$$\mathbb{P}(y \succ \pi^{(t)} | x) \approx \mathbb{P}(y \succ \hat{\pi}^{(K)} | x) := \sum_{k=1}^K \mathbb{P}(y \succ y_k | x) \text{ for some } y_1, \dots, y_K \sim \pi^{(t)}(\cdot | x)$$

$$Z^{(t)}(x) \approx Z^{(t,K)}(x) := \mathbb{E}_{y \sim \pi^{(t)}(\cdot | x)} \left[e^{\eta \mathbb{P}(y \succ \hat{\pi}^{(K)} | x)} \right]$$

Duality gap

$$\max_{\pi} \mathbb{P}(\pi \succ \bar{\pi}_T) - \min_{\pi} \mathbb{P}(\bar{\pi}_T \succ \pi) \leq \mathcal{O}\left(\frac{1}{\sqrt{T}}\right)$$

$$\bar{\pi}_T = \frac{1}{T} \sum_{t=1}^T \pi^{(t)} - \text{mixture policy}$$

proof: arXiv:2405.00675

■ APPROXIMATE NASH EQUILIBRIUM

$$\max_{\pi} \mathbb{P}(\pi \succ \bar{\pi}_T) \geq \mathbb{P}(\pi \succ \bar{\pi}_T) \text{ for all } \pi$$

$$\min_{\pi} \mathbb{P}(\bar{\pi}_T \succ \pi) \leq \mathbb{P}(\bar{\pi}_T \succ \pi) \text{ for all } \pi$$

$$\begin{aligned} \Rightarrow \mathbb{P}(\pi \succ \bar{\pi}_T) &\leq \max_{\pi} \mathbb{P}(\pi \succ \bar{\pi}_T) \leq \min_{\pi} \mathbb{P}(\bar{\pi}_T \succ \pi) + \mathcal{O}\left(\frac{1}{\sqrt{T}}\right) \\ &\leq \mathbb{P}(\bar{\pi}_T \succ \pi) + \mathcal{O}\left(\frac{1}{\sqrt{T}}\right) \end{aligned}$$

Connection to policy gradient

■ POLICY GRADIENT

$$\begin{aligned} & \nabla_{\theta} \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} [r(x, y)] - \frac{1}{\eta} D_{\text{KL}}(\pi_{\theta}(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)) \right] \\ &= \nabla_{\theta} \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} \left[r(x, y) - \frac{1}{\eta} \log \frac{\pi_{\theta}(y | x)}{\pi_{\text{ref}}(y | x)} \right] \right] \\ &= \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} \left[\left(r(x, y) - \frac{1}{\eta} \log \frac{\pi_{\theta}(y | x)}{\pi_{\text{ref}}(y | x)} - b(x) \right) \nabla_{\theta} \log \pi_{\theta}(y | x) \right] \right] \\ &= \eta \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} \left[- \nabla_{\theta} \left(r(x, y) - \frac{1}{\eta} \log \frac{\pi_{\theta}(y | x)}{\pi_{\text{ref}}(y | x)} - b(x) \right)^2 \right] \right] \end{aligned}$$

★ an adaptive policy gradient

$$\theta^{(t+1)}(\cdot | x) = \underset{\theta}{\operatorname{argmin}} \mathbb{E}_{x, y \sim \pi^{(t)}(\cdot | x)} \left(\eta \mathbb{P}(y \succ \pi^{(t)} | x) - \log \frac{\pi_{\theta}(y | x)}{\pi^{(t)}(y | x)} - \log Z^{(t)}(x) \right)^2$$

Summary

■ RECAP

■ GENERAL PREFERENCE

Two-player game

■ NASH RLHF

Mirror-descent self-play, connection to policy gradient